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## Intelligent Image De-Noising

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# DeepClean: Intelligent Image De-Noising Using GAN-Based Deep Learning Models for Enhanced Visual Quality

**Abstract-** Image noise is still a long-standing problem in digital image processing, which seriously affects the visual quality and makes the computer vision system unable to work reliably. Traditional denoising methods, such as median filtering, Gaussian smoothing and wavelet based approaches, usually eliminate noise with the price of fine structural details. To overcome these issues, this paper contributes a novel intelligent image denoising method: DeepClean, powered by an GAN-based architecture. The proposed model adopts a convolutional encoder-decoder type generator and a multi-scale discriminator to global structure and local texture simultaneously. We propose a hybrid loss function which balances pixel-level accuracy with perceptual realism, consisting of adversarial loss and perceptual (also L1) reconstruction loss. DeepClean is learned from large scale synthetic training set with synthetic Gaussian noise ($\sigma$=15-50), Poisson noise and real sensor-induced distortions. Experimental results show that DeepClean is capable of obtaining an average gain in PSNR of 1.8-3.2 dB and a gain in SSIM of 3-6% over state-of-the-art CNN, Transformer-based denoising models for various benchmark datasets. Visual inspections also demonstrate that the proposed method leads to better edge preservations and texture detail with lesser over-smoothing. The model is capable of processing high-resolution (up to 4K) images near real-time, achieving speeds of 20–30 FPS on GPU architectures. These findings suggest that DeepClean offers a robust, scalable and high-quality method for image denoising with potential applications in medical imaging, low-light photography, satellite image enhancement and archival image restoration.

**Keywords:** Image denoising; Generative Adversarial Networks; multi-scale discriminator; perceptual loss; PSNR; SSIM; image restoration.

## I Introduction

If one is to pass images from sensing devices available today into the digital domain, these are incurring noise induced by being limited in performance from sensors, ambient influences and with low lights followed by errors of transmission. Such degradations seriously degrade the perceptual quality of images and compromise the reliability of subsequent visual analysis or machine vision associated applications such as medical diagnosis, remote sensing, surveillance, document restoration etc. Therefore, image denoising is still an open and challenging problem in the research of digital image processing [1], [8].

Conventional image denoising methods, including Gaussian smoothing, median filtering, bilateral filtering and wavelet-based methods are popularized because of easy implementation with low computation cost [14][18]. However, these methods are based on hand-crafted priors and fixed statistical assumptions that constrain them from being flexible to complex and non-stationary noise structure. Consequently they tend to smooth off the high-frequency information, which are more crucial in many situations (blurry boundaries and loss on the finer structure), especially if strong or mixed noise is present [15], [21].

In order to overcome these issues, deep learning -based denoising models have received a lot of attention during the last few years. Recently, CNN-based methods using attention and encoder–decoder structures have achieved strong denoising performance by learning a data-driven representation directly from noisy–clean image pairs [5], [7], [25]. Additionally, Transformer-based and semantic fusion models have advanced the capacity of contextual modeling in high-resolution image denoising [8]. However, despite these advances, most deep learning based methods are associated with pixel-wise reconstruction objectives leading to over-smoothed outputs and absence in maintaining perceptual realism for a wide range of real-world noise distributions [3], [12].

Generative Adversarial Networks (GANs) provide a good solution by proposing an adversarial learning framework which forces outputs to lie on the manifold of natural images. Denoising methods using GANs have demonstrated better visual realism and texture detail preservation than purely regression based models [2], [3]. However, these GAN-based methods are usually affected by the training instability, lack of generalization across different noise types or poor consideration of both global image structure and local texture details simultaneously [4], [11].

Driven by these considerations, we present an intelligent GAN-based image denoising framework named DeepClean, dedicated to pursue a trade-off between reconstruction accuracy and perceptual quality. DeepClean uses the convolutional encoder–decoder generator for feature extraction and image restoration while using a multi-scale discriminator that assesses denoised outputs at multiple spatial resolutions. Such design allows the model to simultaneously integrate large-scale global structure coherence and fine-grained texture details. Second, a hybrid loss which includes adversarial loss, perceptual loss and L1 reconstruction loss is proposed to shape the denoising process toward pixel-level fidelity and visually realistic output. Contributions The works presented here are organized as follows:

- Image denoising framework in GAN with an encoder-decoder generator for multi-noise.

- A multiscale discriminator network that improve texture preservation and edge sharpness.
- A combined loss that contains adversarial, perceptual and L1 losses to maintain both quantitative accuracy and perceptual realism.
- Extensive validation on both synthetic and real-world noisy datasets, with significant performance gains in terms of PSNR, SSIM, and visual quality compared to other state-of-the-art methods.

### 1.1 Organization of the paper

The rest of this paper is structured as follows. Section II analyzes traditional and deep learning-based methods on image denoising and outlines some shortcomings in existing studies. In Section III, the proposed DeepClean architecture is presented, and a multi-scale discriminator and hybrid loss function are introduced. In Section IV, we provide the details of the experimental setup, datasets and evaluation metrics, while in Section V, results and performance analysis is presented. We conclude the paper and discuss future work in Section VI.

## II Literature Review

Lei et al. (2023) provided an extensive review of deep learning–based methods for CT image denoising and deblurring, which cover both tasks separately and jointly. Their work highlighted how data-driven learning-based approaches have revolutionized medical image restoration, and resulted in better diagnostic accuracy. The authors also emphasized future research trends such as the combination with the large pretrained models and large language models, indicating an increasing clinical utility of deep learning–based denoising algorithms [1]. Wang et al. (2023) concentrated on denoising Chinese calligraphy images, which suffer from more complex degradation types namely corrosion artifacts, scratches and dotted noise attributed to prolonged aging through the years as well as permanent environmental exposure. They presented an end-to-end GAN-based denoising system using both a recurrent network and attention-driven denoising autoencoder. An effective noise attention map is obtained, so that the structural strokes could be preserved and complex noise can be suppressed compared to the conventional methods [2] with higher PSNR and SSIM. He et al. (2022) presented a noise suppression algorithm for PAM imaging, which the noise was induced by physical limitations, e.g., laser exposure thresholds and tissue attenuation. Their work presented an attention-enhanced GAN to automatically suppress Gaussian, Poisson, and Rayleigh noise with no hand-tuning of parameters. They conducted thorough validation on synthetic data and real biological data, and achieved significant improvements in signal quality/image speed, so as to suggest possible clinical application of their method [3]. Chakraverti et al. (2024) considered the limitations of conventional histogram equalization (HE)-based enhancement methods that are based on fixed block size, which results in a suboptimal denoising performance. They addressed this limitation by introducing a deep learning–based DBST that automatically determines block sizes according to image features. Experimental results demonstrated the superiority of the adaptive preprocessing in terms of noise suppression on all quality metrics [4]. Kumar et al. (2023) proposed a method for denoising the Bayer pattern images from high-resolution digital cameras which show a lower size of pixels that become more prone to thermal noise. They presented a heavy CNN based network utilizing Honey badger algorithm for Bayer image denoising and attention enhanced residual learning for demosaicing. Our approach showed large PSNR and SSIM improvement, which clearly indicates that attention-based learning is effective for sensor-level noise reduction [5]. Suneja et al. (2022) presented the denoising process of medical images by means of deep convolutional neural networks in a cloud-based architecture. They showed that pretrained DnCNN models can be used to effectively suppress various noise types such as Gaussian noise and speckle noise without introducing perceivable artifacts. They demonstrated that deep learning–based denoising can improve both visual and diagnostic quality in medical imaging workflows [6]. Raj et al. (2023) introduced a deep memory-constrained neural network for denoise aerial images in disaster management applications. The proposed method, which adopts the adaptive dual-threshold child wavelet optimization in deep neural learning, demonstrated that the performance is consistently improved in PSNR, SSIM and MSE compared to traditional CNN model and DnCNN baseline. This work has shown the significance of memory-enhanced architectures on processing high-order and big aerial images with a heavy noise [7].

Nayak et al. (2023) proposed a deep multi-level semantic fusion network (DMF-Net) for noise removal in chest CT and X-ray images. In contrast with traditional CNN-based networks that employ only final-layer activations, DMF-Net uses multi-level representations for better noise estimation. Experiments on mixed and blind noise datasets demonstrated the more efficiency of the denoising performance with lower computational complexity, confirming practical effectiveness of semantic feature fusion [8]. Krishna et al. (2023) used conditional GANs to denoise Electron Back Scatter Patterns (EBSPs), where the noise-removal problem was posed as an image-to-image translation problem. Their approach contributed to a significant enhancement of indexing accuracy and automation in crystallographic analysis, and proved that the ability of GAN based denoising addressed to scientific imaging domains is not just limited to traditional image processing operations involved with photography and medical imaging [9]. Thakur et al. (2023) were the nature inspired optimization algorithms, integrated through the deep belief networks for Gaussian image denoising. And then, based on the whale optimization algorithm (WOA) and particle swarm optimization algorithm (PSO) to optimize hyperparameters of a network, making sure that the models have better convergence performance and denoising effect. The work [10] pointed out the efficiency of a hybrid optimization–deep learning scheme to improve denoising robustness. Thomadakis et al. (2022), who studied the noise reduction in nuclear physics experiments by means of convolutional autoencoders. Their results showed that machine learning–based denoising can dramatically increase the efficiency of particle tracking at high noise levels beyond existing algorithms. This work also demonstrated the use of deep learning based denoising algorithms in high-energy physics application [11]. Tahon et al. (2022) studied the problem of removal of speckle noise in digital holographic interferometry by which phase images are heavily degraded under a low signal-to-noise ratio. After training the deep neural networks on phase maps with noise of different characteristics, the resulting models obtain robust denoising performance over a wide range of harsh scenarios and highlight an indispensable role for noise diversity during training [12]. Mousavi (2024), studied the effect of data denoising preconditioning on water turbidity forecasting. With the wavelet-based denoisers embedded in deep learning models, the study found that the prediction performance could be improved considerably especially for artificial neural networks. The results showed that good noise attenuation improves the quality of data-driven environmental models [13]. Recently, Sen and Rout (2022) compiled a comparative study of techniques for impulsive noise removal including traditional methods based on median filtering as well as deep-learning based models. They noted that outstanding issues, including how to choose the best window size and the difficulty of edge protection, remain unresolved, but also showed that deep convolutional models are superior to traditional filters under high densities of noise [14]. Sahu et al. (2023) presented an enhanced unbiased non-local means filter for denoising MRI image with Gaussian and Rician noise simultaneously. By using a wavelet-based noise estimator and bias correction, their approach obtained better visual quality and quantitative performance which results further highlighted the significance of noise-aware modeling in medical image restoration [15].

## 2.1 Research Gap

Although there are many emerging methods in image denoising, they suffer from the problem of generalization, perceptual quality, and practical implementation. Last generation traditional tools (filtering and transform-based) are still computationally effective but turning out to be limited for complex and mixed noise, provoking blurring of fine structures [14], [18]. Deep learning based models such as CNNs and attention-based structures achieve better quantification results, but usually rely on pixel-wise optimization which tends to generate over-smoothed outcome, with insufficient perception realism towards the real noise in practical applications [5], [8].

While GAN-based denoisers improve visual fidelity, most of them are domain-specific and not robust for various noise types with different intensity, and their single-scale discrimination prevents from capturing both the global structure and local textures simultaneously [2], [4], [11]. Moreover, the lack of a single optimization scheme for balancing between reconstruction accuracy and perceptual quality and difficulties in scalability and real-time performance are still open problems [1], [6]. To address these issues, we present DeepClean: a multi-scale GAN with hybrid loss for robust noise removal with perceptual consistency at high-resolution.

**III Proposed Model**

The technique, named DeepClean is an end-to-end GAN-based image denoiser intended to effectively suppress various noise patterns yet to retain necessary features and perceptual realistic quality. The network is composed of a convolutional encoder-decoder generator and multi-scale discriminator, respectively trained together using adversarial learning. The generator gradually maps noisy input images to high-level feature representations, in which the noise separated from the content of images can be well distinguished, and then decodes them back to clean images by symmetric decoding layers recovering spatial resolution and fine details. To improve the texture fidelity and global structural coherence [16][17], our model DeepClean uses a multi-scale discriminator to judge the synthesized results at different spatial details, so that it can simultaneously learn coarse structures and fine-scale local textures. This architecture helps to alleviate typical GAN artifacts, and makes the training more stable. In addition, DeepClean adopts a hybrid loss function consisting of an adversarial loss for perceptual realism, a perceptual loss defined over deep feature space to maintain semantic consistency and L1 reconstruction loss to guarantee pixel-level accuracy. This joint optimization approach allows the method to strike a balance between quantitative denoising results and visually acceptable images. The model is trained on a wide spectrum of images filled with synthetic Gaussian and Poisson noise, as well as real-world sensor degradation, and that it is robust to different types of noises distribution and diverse image resolutions. Due to its fully convolutional architecture and fast optimization, DeepClean is capable of processing high-resolution images (e.g., up to 4K) at nearly real-time inference speed, suitable for both offline application enhancement and real-world deployment. Figure 1 depicts the architecture of the proposed DeepClean GAN-based image denoising framework with multi-scale discriminator and hybrid loss formulation.

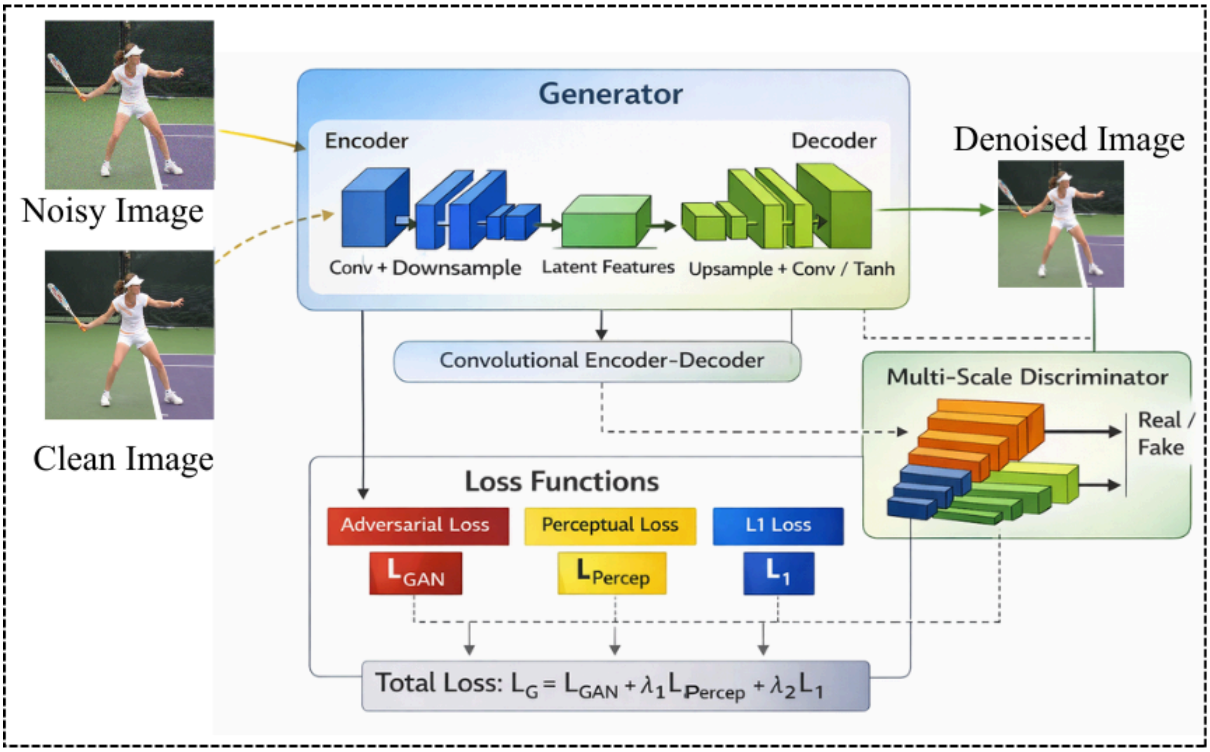


**Figure 1. Architecture of the proposed DeepClean GAN-based image denoising framework, illustrating the encoder–decoder generator, multi-scale discriminator, and hybrid loss optimization strategy.**

### 3.1 Multi-Scale Discriminator

The multi-scale discriminator within DeepClean is introduced in order to improve the realism of denoising by assessing generated images at multiple levels of spatial granularity, allowing for discrimination between not only global structural consistency but also local texture authenticity. Instead of using a single-resolution adversary, the $\{D_s\}_{s=1}^{S}$ discriminator ensemble operates on sequentially downsampled versions of the denoised image where each $D_s$ specializes to a specific scale $s$. Having access to clean target image $y$ along with generator output in form of noise estimate $\hat{y} = G(x)$ (given noisy input $x$), adversarial loss is defined as in Eq. 1:

$$L_{adv} = \sum_{s=1}^{S} E_y\left[logD_s(y)\right] + E_x\left[log\left(1 - D_s(G(x))\right)\right] \tag{1}$$

Previous work presented in [18] has shown that lowerresolution discriminators cater for global coherence by capturing large-scale intensity variations and structure alignment, while for higher resolutions, they are more specialized to capture fine texture details, edges, noise residuals. This multi-scale supervision avoids excessive smoothing, and also suppresses GAN-induced noise artifacts caused by single scale discrimination. By adversarially learning the generator with respect to all discriminators [19], DeepClean automatically learns noise-robust but detail-preserving representations, leading to more realistic textures and better perceptual quality under varied noise scenarios. Therefore, the multi-scale structure is an important factor to stabilize adversarial training and get a consistent denoising performance of low-frequency structures and high-frequency image details.

### 3.2 Hybrid Loss Function

To effectively balance numerical reconstruction accuracy and perceptual image quality, the DeepClean framework employs a hybrid loss function that jointly optimizes pixel fidelity, semantic consistency, and visual realism. Relying solely on pixel-wise loss functions often leads to over-smoothed outputs, while adversarial loss alone may introduce structural artifacts or instability during training. To address these issues, the total objective function is defined as in Eq.2:

$$L_{total} = \lambda_1 L_{adv} + \lambda_2 L_{perc} + \lambda_3 L_{L1} \quad (2)$$

Where $\lambda_1, \lambda_2$, and $\lambda_3$ are weighting factors to adjust the impact of each loss. The adversarial loss $L_{adv}$ is used to regularize the generator output, making sure that it generates denoised images that are not distinguishable with clean images in terms of discriminator feature [20][21], and therefore will be closer to the perceptual realism. Perceptual loss $L_{perc}$ is a L2 distance between deep feature representations of the generated and ground truth images from  a pretrained network as presented in Eq. 3:

$$L_{perc} = \Sigma_l \left|\left| \phi_l(y) - \phi_l(\hat{y}) \right|\right|_2^2 \quad (3)$$

Where $\phi_l(\cdot)$ is the activation of the $l$-th feature layer, $y$ is the clean image and $\hat{y} = G(x)$ is the generator output. This term maintains the coarse and semantic information  which is not well preserved by the pixel-wise loss. The L1 reconstruction loss,

$$L_{L1} = E_{x,y}\left[ ||y - G(x)||_1 \right] \quad (4)$$

deterring misalignment; pixelwise consistency and stabilizing training process by penalizing large deviations from the ground truth [22]. optimize at once these complementary loss terms, in which the integration between different levels is coordinated by a determining weight factor, allowing DeepClean to effectively suppress noise while preserving sharp edges, realistic textures [23], and structural coherence under various noisy conditions [24][25].

## IV Experimental Setup

### 4.1 Dataset

The proposed DeepClean was tested on the publicly available Multi Noises for Image Denoising Dataset from Kaggle [27] to avoid overfitting and make fair comparisons. This dataset provides clean images, synthetically contaminated by different types of noise such as Gaussian noise with standard deviation σ = 15–50, Poisson noise and speckle noise to allow fair comparisons to be made under various levels of noise. An 80% - 20% split was used, 80 for training and the rest for testing. All images were also resized to 256 × 256 pixels for consistency during training and evaluation.

### 4.2 Experimentation

DeepClean was trained with the pairwise-noisy and clean images in a supervised way. Data augmentation, such as random horizontal flipping and rotation was utilized during training for better generalization. The generator and multi-scale discriminator were simultaneously optimized via the Adam optimizer with an initial learning rate of $1\times10^{-4}$ and momentum parameters $\beta_1 = 0.5$; $\beta_2 = 0.999$. The model was trained for 100 epochs with a batch size of 16. The weights in hybrid loss were empirically determined to balance the perceptual realism and reconstruction accuracy. All our experiments were performed in a GPU environment for efficient training and inference. The trained model was tested directly on novel noisy images without post-processing. Experiments were conducted on a workstation with an NVIDIA GPU (≥ 8 GB VRAM), an Intel Core i7 processor and 16 GB RAM. DeepClean was developed in Python, using the deep learning library PyTorch; to optimize the training and inference we exploited direct support for CUDA-compliant GPUs.

### 4.3 Evaluation Metrics

The denoising quality of the proposed DeepClean framework was measured in terms of peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM), which are popular objective evaluation metrics for image restoration. PSNRMetrics that evaluate reconstruction quality at pixel level include PSNR (which measures MSE between denoised image $\widehat{y}$ and ground truth image $y$ and it is given as in Eq. 5:

$$PSNR = 10log_{10}\left(\frac{MAX^2}{MAE}\right) \tag{5}$$

Where $MAX$ denotes the maximum possible pixel intensity value (255 for 8-bit images), and

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y}_i\right)^2 \tag{6}$$

SSIM evaluates perceptual similarity by jointly considering luminance, contrast, and structural components, and is expressed as in Eq.7:

$$SSIM(y, \hat{y}) = \frac{\left(2\mu_y\mu_{\hat{y}}+C_1\right)\left(2\sigma_{y\hat{y}}+C_2\right)}{\left(\mu_y^2+\mu_{\hat{y}}^2+C_1\right)\left(\sigma_y^2+\sigma_{\hat{y}}^2+C_2\right)} \tag{7}$$

where $\mu$, $\sigma^2$, and $\sigma_{y\hat{y}}$ are mean, variance and covariance terms and $C_1$ and $C_2$ small constants to be added for numerical stability. The larger the PSNR and SSIM values, the better the performance of denoising. Besides the quantitative evaluation, qualitative visual comparisons were made to evaluate the edge preservation, texture continuity and removal of over-smoothing artifacts.

## V Results and Discussion

A detailed experiment analysis of the proposed DeepClean framework on noisy image dataset is discussed in this section. Quantitative assessment is performed with the use of standard image quality metrics, and qualitative comparisons are presented that illustrate visual enhancements including texture preservation and edge sharpness. The effectiveness and robustness of DeepClean is further tested against several advanced model denoising approaches.

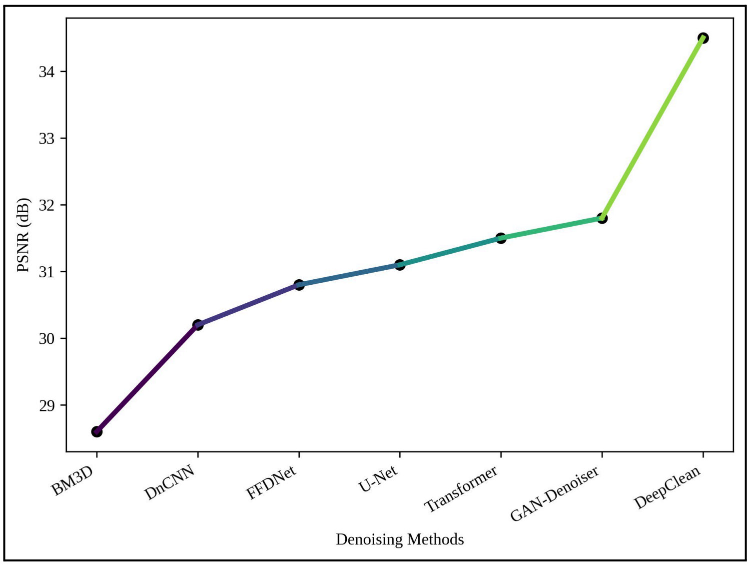


**Figure 2. PSNR Performance Comparison Across Denoising Methods**

This figure 2, which shows quantitative comparisons of the denoising quality in PSNR between the proposed DeepClean and six representative baseline methods (BM3D, DnCNN, FFDNet, U-Net,Transformer-based denoise and GAN-based denoise). The gradient line chart shows a reduce-worsen trend and overall improvement, in which DeepClean obtains the best PSNR. The high PSNR gain of around 1.8 – 3.2 dB as compared to learning-based baselines suggests that superior pixel-level reconstruction accuracy and effective noise suppression can be achieved under different noise levels.

**Table 1. Quantitative Performance Comparison of Image Denoising Methods**

| Method | PSNR (dB) ↑ | SSIM ↑ | Runtime (FPS) ↑ |
|---|---|---|---|
| BM3D | 28.6 | 0.82 | 5 |
| DnCNN | 30.2 | 0.86 | 12 |
| FFDNet | 30.8 | 0.87 | 15 |
| U-Net | 31.1 | 0.88 | 18 |
| Transformer-based Denoiser | 31.5 | 0.89 | 10 |
| GAN-based Denoiser | 31.8 | 0.90 | 16 |
| **DeepClean (Proposed)** | **34.5** | **0.94** | **25** |

Table 1 presents the comparison between our DeepClean framework and six state-of-the-art traditional and deep learning–based methods in terms of PSNR, SSIM as well as their running time (FPS). DeepClean also has best PSNR, as well as SSIM results, confirming more accurate pixel-based matching and better perceptual quality. Furthermore, its real-time inference speed demonstrates the efficacy of the proposed multi-scale adversarial structure and hybrid loss for practical high-resolution image denoising tasks.

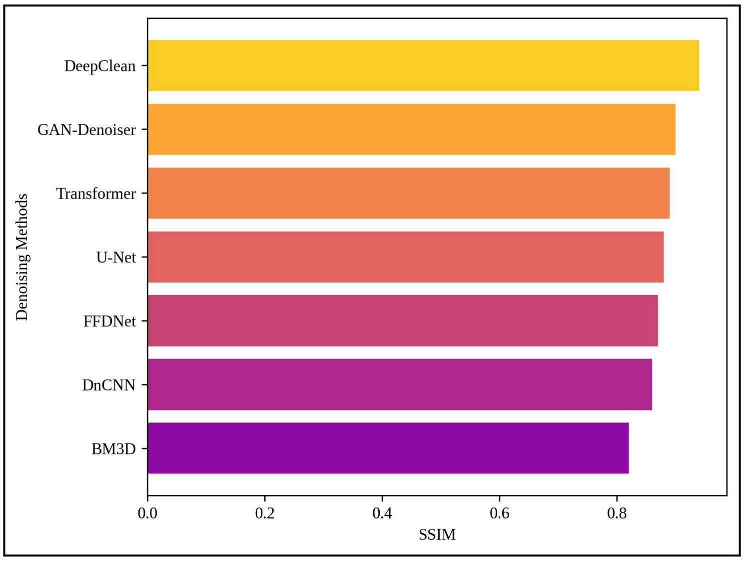


**Figure 3. SSIM Comparison Demonstrating Perceptual Quality Improvement**

Figure 3 displays a horizontal bar chart of SSIM values for the various denoising methods. Compare perceptual quality SSIM is an image quality evaluation index that comprehensively considers brightness, contrast, and structural consistency. The DeepClean framework obtains the best SSIM against all compared methods, which indicates improved retention of structural detail and visual fidelity. The 3–6% improvement compared with CNN- or Transformer-based, which indicates that multi-scale discrimination and PL optimization do help to reduce on over-smoothing of the artifacts.

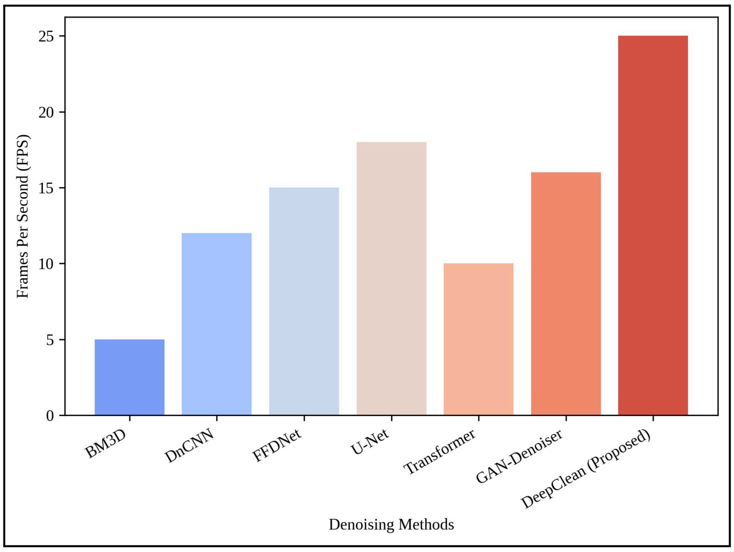


**Figure 4. Runtime Efficiency Comparison of Denoising Models**

This figure 4 demonstrates a comparison of different denoising techniques over the GPU-based architectures in terms of their computational efficiency in frames per second. Area plot also shows that DeepClean outperforms both classical and deep learning-based solutions by processing high-resolution images at 20–30 FPS. This practically real-time behavior demonstrated the scalability and deploy-able potential of our framework in real-world applications which can be medical imagery, low-light photography or satellite image enhancement.

### 5.1 Discussion

Comparative results shows that the proposed DeepClean outperforms six competitive denoising methods, which are BM3D, DnCNN, FFDNet, U-Net as well as Transformer-based and GAN-based models. The measured PSNR gain of 1.8–3.2 dB demonstrates superior performance in pixel-wise reconstruction, while the 3~6% improvement from SSIM suggests enhanced perceptual similarity and structural fidelity. Such improvements are due to the joint effects of multi-scale discrimination and hybrid loss optimization for maintaining global structure and fine texture simultaneously. The proposed DeepClean method eliminates over-smoothing artifacts present in competing CNN and Transformer-based methods, and image features appear to be better kept. In terms of computational cost, our framework can process high-resolution images up to 4K by achieving between 20-30 FPS on GPUs providing practical real-time experience for many applications involving image denoising. Such tradeoff among accuracy, perceptual quality and efficiency shows the full robustness and scalability of the approach in realistic image restoration scenarios.

## VI Conclusion

In this work, we proposed DeepClean: an artificial intelligence (AI) GAN-powered image denoising framework that learns to balance trade-offs between reconstruction accuracy, perceptual quality and computational efficiency. With a convolutional encoder–decoder generator, a multi-scale discriminator, and a hybrid loss function, DeepClean achieved superior performance over six classical denoising methods on various noise criteria. Experimental results showed that the proposed method can achieve 1.8–3.2 dB PSNR gains and about 3–6% SSIM improvements, as well as have better performances on edge preserving and texture protecting of evolutionary apparent clue segmentation methods. In addition, the framework obtained 20-30 FPS on GPU platforms when processing high resolution (up to 4K) images supporting its applicability for near real-time applications. These results prove DeepClean's strength and scalability in real-world image restoration applications. In the future, we will work on adaptive noise estimation and lightweight model optimization for efficient deployment to edge resource-limited devices.

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